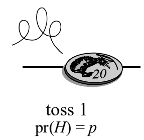
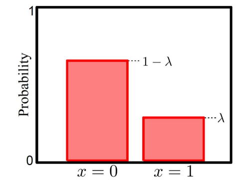
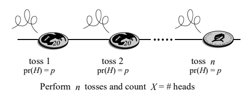
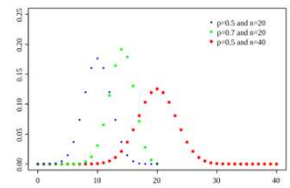
Agenda









- Last time
  - Some distributions used over and over again
  - Bernoulli, Binomial, Geometric
- We capture the mean and variance of these distributions
  - Example of US presidents and the variance
- Today
  - Moment generation functions
  - Hypergeometric

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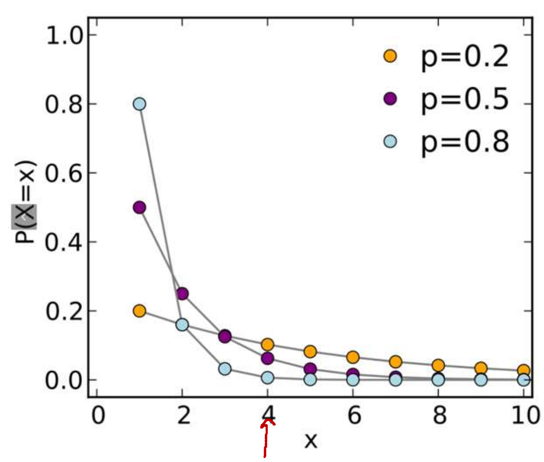
Geometric Distribution

$$P_X(k) = \begin{cases} p(1-p)^{k-1} & \text{for } k = 1, 2, 3, \ldots \\ 0 & \text{otherwise} \end{cases}$$

X = k

- X ~ Geom(p)
- A sequence of independent Bernoulli trials until you observe the first success
- "Memoryless"



- Octave: geopdf(3, 0.2)
- from scipy.stats import geom
  geom.pmf(4,0.2)

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Geometric Distribution

$$P_X(k) = \begin{cases} p(1-p)^{k-1} & \text{for } k = 1, 2, 3, \ldots \\ 0 & \text{otherwise} \end{cases}$$

- X ~ Geom(p)
- A sequence of independent Bernoulli trials until you observe the first success
- E[X] = ?
  - E[X] = 1.p + (1-p)[1 +E[X]]
  - E[X] = 1/p
- Var(X) = $(1-p)/p^2$

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Moment generating function

- The moment generating function (MGF) of X is

May not exist

$t=0$

$$\phi_X(t) = E(e^{tX}) = \sum_x e^{tx} P(X = x) \quad \text{(discrete r.v.)}$$

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Why MGF?

- Observe differentiating the MGF with respect to t

$$\phi_X(t) = E(e^{tX})$$

$$\phi_X{}'(t) = \frac{d}{dt}(E(e^{tX})) = E\left(\frac{d}{dt}(e^{tX})\right) = E(Xe^{tX})$$

$$\phi_X{}'(0) = E(X)$$

$$\phi_X^{(2)}(t) = \frac{d}{dt}(E(Xe^{tX})) = E(X^2e^{tX})$$

$$\phi_X^{(2)}(0) = E(X^2),\ \dots,\ \phi_X^{(n)}(0) = E(X^n)$$

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Geometric Distribution

$$P_X(k) = \begin{cases} p(1-p)^{k-1} & \text{for } k = 1, 2, 3, \dots \\ 0 & \text{otherwise} \end{cases}$$

$$M_X(t) = E[e^{tX}] \quad = \sum_{k=1}^{\infty} e^{tk} P_X(k)$$

$$M_X(t) = \sum_{k=1}^{\infty} e^{tk} pq^{k-1}$$

$$q = 1 - p$$

$$M_X(t) = pe^t \sum_{k=1}^{\infty} (qe^t)^{k-1}$$

$$= pe^t \sum_{j=0}^{\infty} (qe^t)^j$$

$$= \frac{pe^t}{1 - qe^t}.$$

$$M_X'(t) = \frac{pe^t(1 - qe^t) - pe^t(-qe^t)}{(1 - qe^t)^2}$$

$$= \frac{pe^t - pqe^{2t} + pqe^{2t}}{(1 - qe^t)^2}$$

$$= \frac{pe^t}{(1 - qe^t)^2}.$$

$$E[X] = M_X'(0)$$

$$= \frac{p}{(1-q)^2}$$

$$= \frac{1}{p}.$$

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Geometric Distribution

$$P_X(k) = \begin{cases} p(1-p)^{k-1} & \text{for } k = 1, 2, 3, \ldots \\ 0 & \text{otherwise} \end{cases}$$

$$q = 1 - p$$

$$\begin{aligned} M_X(t) &= E[e^{tX}] \\ &= \sum_{k=1}^{\infty} e^{tk} p q^{k-1} \\ &= pe^t \sum_{k=1}^{\infty} (qe^t)^{k-1} \\ &= pe^t \sum_{j=0}^{\infty} (qe^t)^j \\ &= \frac{pe^t}{1 - qe^t}. \end{aligned}$$

$$\begin{aligned} M_X''(t) &= pe^t(1-qe^t)^{-2} + pe^t\left[-2(1-qe^t)^{-3}(-qe^t)\right] \\ &= \frac{pe^t}{(1-qe^t)^2} + \frac{2pqe^{2t}}{(1-qe^t)^3}. \end{aligned}$$

$$\begin{aligned} E[X^2] &= M_X''(0) \\ &= \frac{p}{(1-q)^2} + \frac{2pq}{(1-q)^3} \\ &= \frac{1}{p} + \frac{2q}{p^2}. \end{aligned}$$

$$\begin{aligned} \mathrm{Var}(X) &= E[X^2] - E[X]^2 \\ &= \frac{1}{p} + \frac{2q}{p^2} - \frac{1}{p^2} \\ &= \frac{p + 2q - 1}{p^2} \\ &= \frac{q}{p^2} \\ &= \frac{1-p}{p^2}. \end{aligned}$$

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Moment

- The moment generating function (MGF) of X is
- The moment of X of order n is
  - The first moment is the mean
- The central moment of X is E[(X – E[X])n]
  - The second central moment is the variance

$$m_{\mathrm{n}} = E(X^{\mathrm{n}}),\ \mathrm{n} \geq 1$$

$$\phi_X(t) = E(e^{tX}) = \sum_x e^{tx} P(X = x) \quad \text{(discrete r.v.)}$$

$$= \int_{-\infty}^{\infty} f_X(x)\, e^{tx}\, dx \quad \text{(continuous r.v.)}$$

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Why MGF?

$m_{\mathrm{n}} = E(X^{\mathrm{n}}),\ \mathrm{n} \geq 1$

- Because in the series expansion, every "moment" m_i appears as a coefficient
  - The function generates the moments.

$$e^{tX} = 1 + tX + (tX)^2/2! + (tX)^3/3! + \cdots$$

$$\phi_X(t) = E(e^{tX}) = 1 + t\,m_1 + t^2 m_2/2! + t^3 m_3/3! + \cdots$$

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Uniqueness

- If two discrete random variables X and Y have MGFs $\phi_x(t)$ and $\phi_y(t)$ that both exist and agree for all t, then X and Y have the same pmf
- Same holds for continuous random variables also

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Other properties

- If Y = aX + b $\phi_Y(t) = e^{tb}\,\phi_X(at)$
- If X and Y are independent $\phi_{X+Y}(t) = \phi_X(t)\,\phi_Y(t)$
- If Z equals X with probability p and Y with probability 1 − p: $\phi_Z(t) = p\,\phi_X(t) + (1-p)\,\phi_Y(t)$

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Sampling with and without replacement

- Suppose there are k objects each of a different type
- When you sample two different objects with replacement,
  - You pick a particular object with probability 1/k, and
  - You place it back (replace it)
  - You pick the second object. The probability of picking an object of another type is again 1/k.
- When you sample without replacement,
  - probability that your first object was of so and so type is 1/k
  - the probability that your second object was of so and so type is now 1/(k-1) because you didn't put the first object back

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